A Toy Model of Consciousness as a Maintained Selective Field

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Abstract

This paper presents a toy model of consciousness approached as a maintained selective field rather than as a mere list of contents or a sequence of isolated states. The model is not offered as a quantitatively faithful simulation of human consciousness, nor as a solution to the hard problem. Its aim is more limited and exploratory: to see whether a field-based rendering of consciousness can be formalized in a way that still produces coherent and nontrivial dynamics. The model represents disclosures as weighted tokens distributed across a maintained field and tracks salience, articulation, carriage, uptake, weighting, foreground/background position, residue, closure, fragmentation, and correction. The paper motivates the model, specifies its formal structure, explains its parameters and derived ratios, and examines canonical scenarios and parameter sweeps. The guiding question is whether behaviors such as weak carriage, failed uptake, fixation, premature closure, fragmentation, and correction can be rendered in a reasonably stable and interpretable way rather than appearing only as artifacts of narrow or opaque tuning.

Introduction

Much writing on consciousness tends to gather around two familiar poles. On one side are highly general philosophical treatments that remain suggestive but leave the object underdetermined once one asks what, exactly, is supposed to be organized, carried, revised, or sustained over time. On the other side are empirical or computational treatments that move quickly into mechanisms, tasks, or correlates while presupposing a flatter target: a reportable state, an attentional bottleneck, a global workspace event, a classification problem, or some other locally tractable surrogate. Both approaches can be illuminating. Still, neither by itself quite provides the kind of explicit middle-level object that lends itself well to formal experimentation. What seems worth trying, at least provisionally, is a modelable rendering of the field-like organization through which conscious life is ordinarily lived.

The present paper takes up that narrower and more tentative task. It introduces a toy model of consciousness understood as a maintained selective field. The model is not offered as a quantitatively faithful simulation of human consciousness, as a theory of neural realization, or as a solution to the hard problem. Its aim is structural and exploratory. The question is whether a field-based rendering of consciousness can survive formalization without immediately thinning into trivial bookkeeping or opaque parameter play. More concretely, the model asks whether a small set of interpretable variables can generate coherent and nontrivial dynamics for phenomena that seem central to conscious life: differential salience, foreground and background organization, weak and strong carriage, persistence, residue, closure, fragmentation, and correction. If that proves possible, even at a toy level, then the rendering may be worth keeping around as a useful intermediate object for further criticism and refinement.

The motivating intuition is simple. Consciousness is not especially well captured as a mere list of contents or as a sequence of isolated states. What ordinary life seems to show, at least in the rendering I am trying to preserve here, is an organized field in which different disclosures have different force, different clarity, different persistence, and different access to uptake. Some things stand forward. Some remain active only as background pressure. Some matter before they are clearly articulable. Some structures harden too quickly. Some fail to integrate. Some return until they are either metabolized or fixed into the field more deeply. The aim is not to capture everything, but to preserve enough of this field structure that some of its characteristic organizations and misfires can be rendered dynamically.

Four commitments govern the paper from the outset:

- the model targets *structural fidelity* rather than quantitative fidelity;
- the modeled object is a *maintained selective field*, not a container of contents;
- the model's outputs are interpreted in terms of *regimes* and *failure signatures*, not literal diagnoses;

- the result is meant to be a *portable research artifact*: inspectable, criticizable, revisable, and useful for later refinement.

On that basis, the paper proceeds in four stages. It first clarifies why a toy model may be useful here and what structural features the model is meant to preserve. It then specifies the model itself: architecture, formal structure, update dynamics, parameters, ratios, and operational signatures. It next examines canonical scenarios, baseline runs, sweeps, and visualization strategies, with special attention to subject-position and regime behavior. It closes by assessing what the model seems to get right, where arbitrariness remains, and what claims such a model can and cannot reasonably support. The ambition throughout is intentionally bounded. The paper does not try to prove that consciousness has been captured. It tries to offer one specific rendering of consciousness in a form that can be inspected, criticized, and revised.

Why a Toy Model Is Needed

A toy model seems useful here because the object under discussion is easy to gesture toward and harder to test. Once consciousness is described as a field-like order of differential salience, uneven articulation, background pressure, persistence, and possible correction, the prose can remain persuasive while still leaving major ambiguities in place. Which distinctions are actually load-bearing? Which relations are merely verbal? Which supposed features of the object generate recognizable dynamics, and which do little once formalized? A toy model does not eliminate those questions, but it does place them in a more explicit form. It makes the rendering answerable to construction.

That answerability matters for at least three reasons. First, formalization exposes hidden dependence. A verbal account may speak comfortably of foreground and background, or of closure and correction, without yet specifying how those differ operationally or how they interact over time. A model requires decisions about state, update, coupling, persistence, and threshold. Second, a toy model can help indicate whether the proposed object has any generative grip. If the rendering is doing real work at all, one might expect at least some nontrivial behavior to emerge when its principal dimensions are formalized together. Third, it becomes easier to distinguish robust structure from accidental tuning. A purely verbal account can always redescribe outcomes after the fact. A model at least makes it possible to ask whether the behaviors that matter persist across interpretable parameter ranges or appear only under narrow and opaque settings.

The point of using a toy model here is therefore not to imitate the standards of a mature empirical science prematurely. It is to secure a somewhat stricter discipline than prose alone can provide while remaining clear about the model's limits. The present model is not introduced as a quantitatively faithful simulation of human consciousness. It is not calibrated against neural data, fitted to behavioral datasets, or offered as evidence that machine consciousness has been achieved or approximated in any strong sense. It is a structural toy model. Its purpose is narrower: to see whether consciousness, rendered as a maintained selective field, yields coherent and nontrivial

dynamics when translated into a small formal system. The relevant question is not whether the model gets the numbers right, but whether it preserves enough of the object's organization to generate intelligible regimes.

That means the model should be judged by a different standard than a high-fidelity empirical simulator. The main concern is structural fidelity. A useful toy model should preserve the relations that motivate the rendering in the first place: differential salience, foreground and background organization, weak and strong carriage, persistence across steps, residue as structured remainder, and correction as endogenous reorganization rather than external reset. If those relations disappear under formalization, then this particular field rendering is probably not doing enough work. If they remain visible and give rise to recognizable patterns such as fixation, pressure-heavy under-articulation, fragmentation, closure-heavy narrowing, or reopening through correction, then the rendering has at least shown itself to be worth developing further, even if the current parameter values remain provisional.

There is another reason a toy model is appropriate here. The object in question sits at an awkward level for both purely philosophical and purely empirical treatment. It is too structured to remain only a metaphor, yet too broad and multi-dimensional to be captured well by a single local mechanism. A toy model is useful precisely in such cases. It can occupy the middle range between informal description and overcommitted implementation. It can represent a field without pretending to exhaust the lived phenomenon. It can clarify what kinds of variables and couplings seem indispensable, which kinds of failure signatures belong naturally to the model, and where the current rendering becomes arbitrary or underdetermined. Used this way, the model is not a shortcut around conceptual work. It is one of the ways that conceptual work becomes criticizable.

The result is best understood as a portable research artifact rather than as a finished theory. Its value lies in being inspectable, revisable, and able to support discussion about what the underlying rendering does and does not buy. A good toy model makes criticism easier. It invites one to ask whether the parameters are meaningful, whether the regimes are robust, whether the failure signatures are over-read, whether the scenarios are doing too much of the work, and whether the same behavior could be generated more simply. Those are not objections to modeling. They are part of what makes modeling useful. If the present model can sustain that level of scrutiny while still producing coherent regime behavior, then it has probably done something useful.

Target Intuition: Consciousness as a Maintained Selective Field

The model begins from a simple intuition: consciousness is better approached as a maintained selective field than as a mere inventory of contents or a sequence of isolated states. What matters in ordinary experience is not only *what* is present, but *how* it is present. Some disclosures stand out with force. Some remain in the background while still shaping the field. Some are clear enough for explicit uptake. Others persist only as pressure, tendency, mood, or unresolved significance. A toy model meant to preserve anything recognizably consciousness-like should therefore aim at

organization rather than content-listing alone.

The field rendering used here is meant to preserve a small set of structural features:

- differential salience
- foreground and background organization
- weak and strong carriage
- variable articulation
- persistence across time
- residue as structured remainder
- correction as endogenous reorganization

These are the features the model is designed to keep in play. They do not exhaust consciousness, but without them the target object becomes too flat to be worth formalizing in the present way.

To call consciousness a *field* is to say that conscious life has spread, asymmetry, and internal distribution. At any moment, different disclosures can coexist with different force and different access to uptake. A perception may be focal while a bodily tension remains ambient. A concern may stay in the background while silently altering what later becomes salient. A mood may weight the significance of the whole scene without becoming a sharply bounded object among others. The point is not that consciousness contains many things at once. The point is that those things belong to one unevenly organized order in which some are foregrounded, some are recessive, and some continue to matter without being well articulated.

To call the field *selective* is to reject both arbitrariness and exhaustiveness. Consciousness neither invents what it lives through wholesale nor presents everything that bears upon it in equal detail. It weights, filters, and partially carries what matters. This is why the model includes both stronger and weaker forms of carriage. Not everything that is experientially active is equally available for explicit reflection, report, or action. A subject can live under the pressure of something not yet clearly thinkable. A model that cannot represent that difference may still simulate task performance or attention shifts, but it would leave out something important in the rendering being tested here.

To call the field *maintained* is to emphasize continuity without requiring identity of content. Consciousness persists not by keeping the same perception, thought, or feeling fixed across time, but by sustaining an order in which disclosures can recur, fade, re-enter, compete, and be revised. This is why persistence, decay, propagation, and correction are central to the model. The target is not a snapshot of conscious contents. It is a living organization that can hold a foreground and a background, accumulate residue, harden into closure, fragment under stress, or reopen through

successful correction.

This target intuition should also be read negatively. The model is not trying to capture the full phenomenological richness of conscious life, to reproduce subjective feel in any exhaustive sense, or to settle the metaphysical question of why experience exists at all. It is trying to preserve enough of the field structure that recognizable dynamics become possible. If the resulting model can generate interpretable regimes for weak carriage, fixation, fragmentation, closure, and correction, then the field rendering has at least become more testable than it would be in prose alone. If it cannot, then the intuition remains less well motivated than it may initially sound.

Model Architecture

The model represents consciousness as a maintained field of disclosure-tokens rather than as a flat store of contents or a single global state. A disclosure-token is not meant to be an object in the world itself. It is a locally active unit of conscious organization: something present in the field as perceptual appearance, affective pressure, bodily signal, memory activation, thought-fragment, or reflexive monitoring content. The model therefore begins from a set of tokens whose significance lies not only in their individual state, but in their position within a wider field of weighting, competition, persistence, and uptake.

At the highest level, the architecture has five components:

- a population of disclosure-tokens
- a field of weighted relations among those tokens
- a subject-relative uptake operator
- a foreground/background organization
- a correction mechanism driven by accumulated tension in the field

These components are meant to preserve the minimal structure of a maintained selective field. The tokens supply local disclosures. The relational field allows those disclosures to reinforce, inhibit, or condition one another. The uptake operator marks which parts of the field are actually being inhabited most strongly at a given step. Foreground and background distinguish stronger from weaker carriage. Correction prevents the model from collapsing into a purely feedforward or purely accumulative process by allowing the field to reorganize under pressure.

Each token carries a small state rather than a full symbolic description. The point is not to encode rich semantic content in advance, but to represent a disclosure in terms of a few dimensions that matter structurally for conscious organization. In the current implementation, those dimensions include mode, salience, articulation, carriage, uptake, weighting, and foreground/background position. Mode distinguishes broad kinds of disclosure such as perceptual, affective, cognitive,

bodily, imaginal, mood-toned, or reflexive material. Salience measures how strongly a token claims the field. Articulation measures how structured and revisitable it has become. Carriage tracks whether the token remains sustained enough to be experientially carried rather than merely flashing locally. Uptake records the degree to which the field is currently inhabiting that token. Weighting marks its significance within the field even when it is not fully foregrounded. Foreground/background position records whether the token sits forward, recessively, or in an intermediate state. Affective pressure and orientation still matter in the model, but they are expressed through mode, orientation state, and affective-amplification parameters rather than as separate token fields. None of these dimensions is sufficient on its own. Together they give each token enough structure to participate in a genuinely organized field.

The field itself is not just the set of active tokens. It also includes the weighted relations among them. Some disclosures reinforce one another, some compete, some propagate framing, and some remain only loosely coupled. Those relations are essential because the model is not trying to simulate independent items merely fighting for a single attention slot. It is trying to represent a maintained order in which what is already active can shape what later becomes salient, what gains articulation, what returns after fading, and what becomes difficult to dislodge once established. Field structure is therefore one of the main places where continuity, clustering, fixation, and fragmentation become possible.

The subject-position is represented in the model not as a privileged token but as an uptake operator over the field. This matters conceptually. A subject-position is not here treated as one more piece of content inside consciousness. It is the organized standpoint relative to which some disclosures are taken up more strongly than others. Formally, the model therefore computes uptake as a weighted profile across tokens rather than designating a single token as "the self." This allows the subject-position to be represented as concentrated, diffuse, stable, conflicted, or fragmented depending on the state of the field. It also makes possible later visualizations in which subject-position appears as a moving uptake profile rather than a fixed point.

Foreground and background are implemented as dynamically changing regions of stronger and weaker carriage rather than as hard partitions. Tokens can move toward stronger uptake, recede without disappearing, or remain active in a low-articulation state for many steps. This is where the model begins to depart from simpler architectures that track only what is currently selected. Background activity matters because much of the field's structure depends on what is still present but not focal: lingering concern, weakly carried pressure, ambient tone, partially suppressed significance, or a latent framing that will later reactivate. The model therefore has to keep background matter live enough that recurrence, residue, and delayed articulation become possible.

Finally, the architecture includes a correction pathway. Without such a pathway, the model would be able to accumulate dominant structures but not to test how those structures might be revised from within. Correction in this model is not an external supervisor resetting a bad state from outside. It is an endogenous response to accumulated tension in the field: residue, conflict, fragmentation, or excessive closure can produce pressure for reweighting, reframing, and

reorganization. This is one of the model's more consequential architectural commitments. A maintained selective field that could weight and persist but never reopen would likely be too static for the phenomena the model is meant to explore.

Taken together, these architectural choices define the scope of the model. It is not a global workspace implementation, not a symbolic planner, not a full predictive-processing account, and not a neuroscience simulator. It is a deliberately reduced field architecture designed to ask a narrower question: if consciousness is rendered as a maintained selective field, what is the smallest formal structure that can still generate recognizable dynamics of salience, carriage, persistence, residue, closure, fragmentation, and correction?

Formal Structure

At time t, the model represents consciousness as a structured field

$$\mathcal{C}_t = (C_t, W_t, U_t),$$

where C_t is the active set of disclosure-tokens, W_t is the weighted relational structure among them, and U_t is the subject-relative uptake profile defined over the field. This notation is intentionally modest. It does not claim that consciousness literally *is* a set, a matrix, and an operator. It says only that, for the purposes of the toy model, these are the minimal formal elements needed to preserve the field-character of the object being tested.

Each token $d_i^t \in C_t$ is represented as a small state vector,

$$d_i^t = (m_i,\ s_i^t,\ a_i^t,\ c_i^t,\ u_i^t,\ w_i^t,\ g_i^t),$$

with the following interpretation:

- m_i: mode type
- s_i^t: salience
- a_i^t: articulation
- c_i^t: carriage
- u_i^t: uptake
- w_i^t: weighting
- g_i^t: foreground/background position

The mode variable m_i does not encode full phenomenological richness. It only marks broad

differences in how a disclosure enters and behaves within the field. In the current model,

$$m_i \in \{\text{perc}, \text{aff}, \text{mood}, \text{cog}, \text{imag}, \text{body}, \text{refl}\}.$$

Salience $s_i^t \in [0,1]$ measures how strongly the token claims the field at a given step. Articulation $a_i^t \in [0,1]$ measures how structured and revisitable the disclosure has become. Carriage $c_i^t \in [0,1]$ measures whether the disclosure remains experientially sustained. Uptake $u_i^t \in [0,1]$ measures how strongly the field is currently taking it up. Weighting $w_i^t \in [0,1]$ measures its significance within the current field organization. Foreground/background position $g_i^t \in [-1,1]$ records whether it sits forward, recessively, or between those poles.

The field is not exhausted by the token set. It also includes a weighted relation matrix

$$W_t = (w_{ij}^t),$$

where w_{ij}^t measures how strongly token d_i^t conditions token d_j^t. A positive weight can represent reinforcement, resonance, or propagative support. A negative weight can represent inhibition, competition, or suppression. The matrix need not be symmetric. One token may disproportionately organize another without receiving equivalent support in return. That asymmetry matters for modeling closure, propagated framing, and fixation. The field therefore has internal geometry: not all disclosures stand in the same relation to all others.

Subject-position is represented through an uptake profile rather than a dedicated self-token. Formally, the model defines

$$U_t = \{u_i^t\}_{i=1}^{|C_t|},$$

with each $u_i^t \in [0,1]$ indicating how strongly token d_i^t is currently taken up. In the executable model these uptake values are not globally normalized; they are local token variables. When the notebook derives subject-position geometry, it normalizes uptake weights secondarily in order to compute a center, spread, and orientation vector. Some tokens are taken up strongly, some weakly, and some remain active with little or no direct uptake. That distinction is one of the main reasons uptake is modeled separately from salience. A disclosure may stand out without being well metabolized, and it may remain weakly carried even while continuing to shape the field.

Foreground and background are defined operationally rather than metaphysically. Given a foreground/background threshold θ_F, the foreground is

$$F_t = \{d_i^t \in C_t \mid g_i^t \geq \theta_F\},$$

while the background is

$$B_t = C_t \setminus F_t.$$

This is not meant to imply a hard phenomenological boundary. It is a formal convenience for distinguishing stronger from weaker carriage. The important point is that background tokens

remain in the field. They continue to affect weighting, recurrence, and later uptake rather than disappearing once they fall below focal threshold.

Residue is modeled as structured remainder rather than mere error. The current formalization treats residue pressure as the combination of under-articulated weighted material that remains weakly carried or backgrounded together with unresolved residue traces. One simple form is

$$P_t^{\text{res}} = \sum_{i \in C_t^*} (1 - a_i^t) w_i^t + \sum_k r_k^t,$$

where C_t^* denotes the tokens whose carriage is low or whose foreground/background position remains recessive, and r_k^t denotes unresolved residue traces. This quantity rises when the field contains material that is significant enough to matter but insufficiently articulated or integrated to be carried well. The interpretation is deliberately structural. A high residue state does not mean the model has discovered hidden truth. It means the field is carrying unresolved pressure.

The model also tracks closure and fragmentation as field-level observables. Closure is treated as concentration of uptake around a narrow cluster of disclosures. If

$$\pi_i^t = \frac{u_i^t}{\sum_j u_j^t},$$

then a simple closure index is

$$K_t = \sum_i (\pi_i^t)^2.$$

Higher values indicate tighter concentration and, therefore, greater risk of narrowing or fixation. Fragmentation is tracked through the coherence of the weighted field. One useful form is

$$\Phi_t = 1 - \frac{\sum_{i \neq j} |w_{ij}^t|}{|C_t|(|C_t| - 1)},$$

possibly adjusted by a mode-coupling term. Higher fragmentation means weaker effective integration across the active field.

Correction pressure is then modeled as a compound function of unresolved tension in the field. In the current version,

$$P_t^{\text{corr}} = \lambda_R R_t + \lambda_K K_t + \lambda_\Phi \Phi_t + \lambda_D D_t,$$

where D_t is a conflict or dissociation term and the λ's weight the relative contribution of residue, closure, fragmentation, and mode-level discord. The model does not assume that every kind of tension should immediately trigger revision. It assumes only that some tensions accumulate and can, beyond threshold, drive field reorganization. Correction therefore enters the formal system not as an external override but as an endogenous response to accumulated mismatch.

These definitions are deliberately spare. Their purpose is not to give a complete mathematics of consciousness, but to stabilize a formal vocabulary in which the field can be updated, measured,

and compared across scenarios. What matters most at this stage is that each quantity names a theoretically meaningful aspect of the rendering: tokenized disclosures, relational structure, uptake, foreground/background differentiation, residue, closure, fragmentation, and correction pressure. If the model generates recognizable regimes with only this much formal machinery, then the field intuition has at least shown itself to be formalizable in a nontrivial way.

Update Dynamics

The formal structure just defined is static unless a rule is given for how the field changes from one step to the next. The central question for this section is therefore not what the field *is* at an instant, but how it is maintained as an organized field at all. The model does not treat time as a simple succession of unrelated snapshots. It assumes that what is already active in the field conditions what can become salient next, what can gain articulation, what can persist in the background, and what can eventually trigger revision. The update cycle is meant to preserve that dependence without making the model so complicated that every behavior becomes a tuning artifact.

Each update step has seven main moments:

1. token generation or reactivation
2. salience update
3. articulation update
4. persistence and decay update
5. uptake allocation
6. relational propagation
7. correction and field reorganization

These moments are analytically separable even when, in implementation, some of them are coupled inside the same local update. Their order is not meant to represent a literal introspective sequence. It is a structural device for modeling how a maintained selective field can change without collapsing into either pure feedforward flow or pure state persistence.

The cycle begins with token generation or reactivation. Some tokens enter the field because of current input: a perceptual event, a bodily alarm, a memory trigger, a new cognitive framing, or a reflexive monitoring token. Others do not enter from outside in any strong sense, but reappear because already active structures keep them near threshold. In schematic form, the field at $t+1$ begins from a union of surviving tokens and newly active candidates:

$$C_{t+1}^{\text{raw}} = C_t^{\text{survive}} \cup C_{t+1}^{\text{new}}.$$

This distinction matters because the model is not trying to represent consciousness as a field rebuilt from scratch at each step. Some disclosures are genuinely new. Others recur because the field has not finished with them.

Salience is then updated. A token's salience at the next step depends not just on its current salience, but on its affective weight, the support or inhibition it receives from neighboring tokens, its persistence, and the current regime parameters. In loose form:

$$s_i^{t+1} = f_s\Big(s_i^t,\ w_i^t,\ O_t,\ \sum_j w_{ji}^t s_j^t,\ \Theta\Big).$$

The exact function f_s is not meant to be psychologically exact. What matters is that salience is path-dependent and relational. A disclosure becomes strong not only because it is already weighted, but because it is reinforced by what is already active, modulated by the current orientation state, or supported by a field that has become organized around it. This is where propagated framing and threat-heavy or mood-heavy capture begin to emerge.

Articulation is updated separately. The model assumes that something can be salient without yet being well articulated, and that articulation can rise when a token receives sustained uptake, cross-mode support, or favorable bandwidth conditions. One schematic form is

$$a_i^{t+1} = f_a\Big(a_i^t,\ u_i^t,\ \sum_j w_{ji}^t u_j^t,\ A_b,\ m_i\Big),$$

where A_b is articulation bandwidth. A token may therefore remain vivid yet poorly structured, or may become more revisitable as the field gives it enough support to stabilize. This distinction is one of the main reasons the model can represent pressure-heavy but under-articulated states. If salience and articulation were collapsed into one variable, many of the behaviors the model aims to capture would disappear.

Persistence and decay are updated together. Some disclosures remain active because they are reinforced by uptake and relation structure; others fade because they lose support and are not reactivated. The persistence term p_i^t tracks how much carryover a token can exert, while the decay parameter D limits indefinite accumulation. In simplified form,

$$p_i^{t+1} = f_p\Big(p_i^t,\ u_i^t,\ \sum_j w_{ji}^t u_j^t,\ P,\ D\Big).$$

This is one of the main sites where regime differences become visible. High persistence relative to decay supports recurrence and fixation. Low persistence relative to decay supports thinner fields, faster turnover, and fragmentation-prone organization. The model does not assume that persistence is always pathological or always healthy. It assumes only that persistence is one of the central structural features by which a field ceases to be merely momentary.

Uptake is then allocated across the field. Uptake is modeled as a weighted response to the token

state rather than as a hard winner-take-all selector. A token with high salience may still receive weak uptake if it remains poorly articulated or badly integrated. Conversely, some moderately salient but well-structured token may receive stable uptake because it is easier for the field to inhabit. In schematic terms:

$$u_i^{t+1} = f_u\Big(s_i^{t+1},\ a_i^{t+1},\ c_i^{t+1},\ w_i^{t+1},\ O_t,\ S_t\Big),$$

with uptake remaining a local field variable rather than a hard winner-take-all selector. Uptake is therefore the moment at which subject-position becomes formally visible. The field may be concentrated, diffuse, conflicted, or fractured depending on how uptake distributes itself. This is also one of the main drivers of later foreground/background movement: not all active tokens receive equal habitation.

Relational propagation follows. Once a token has been strongly taken up, it can alter the weights and influences within the field. This is where framing becomes self-reinforcing or where cross-mode support strengthens a disclosure that was previously weakly carried. In the model, W_t is not fixed. It changes as active patterns propagate support, inhibition, or mode-linking:

$$W_{t+1} = f_W\Big(W_t,\ U_{t+1},\ C_{t+1},\ M_c,\ C_w\Big),$$

where M_c is cross-mode coupling and C_w is closure weight. High closure weight strengthens dominant framings and suppresses alternatives. High cross-mode coupling allows perception, affect, thought, bodily registration, and reflexive monitoring to recruit one another more effectively. This step is essential because many conscious distortions are not produced by local token strength alone, but by the way an active structure begins to condition what else can become available.

Correction comes last, not because it is temporally last in lived experience, but because it is modeled as a response to tensions accumulated through the rest of the cycle. If residue, closure, fragmentation, or dissociation rise past threshold, the field does not simply continue unchanged. Instead, a correction operator can reweight relations, reopen suppressed alternatives, or allow weakly carried material to gain stronger uptake. In simple terms:

$$\text{if } P_t^{\text{corr}} > \theta_{\text{corr}}, \qquad \mathcal{C}_{t+1} \leftarrow \Gamma(\mathcal{C}_{t+1}),$$

where Γ is a reorganization operator rather than a reset. This distinction matters. Correction is not external policing. It is endogenous reconfiguration in response to accumulated misfit within the field. A model without this step could generate narrowing and persistence, but it could not represent reopening, metabolization of pressure, or transition out of weak carriage into stronger articulation.

Two design choices should be emphasized. First, the update cycle is intentionally asymmetric. Salience does not simply mirror articulation, and uptake does not simply mirror salience. This is what lets the model distinguish between what stands out and what can actually be worked

with. Second, the cycle is intentionally recursive. Earlier uptake shapes later salience and relation structure, which in turn shape later uptake again. Without that recursion, closure, fixation, recurrence, and correction would all be much harder to model in a way that belongs to the field rather than being imposed from outside.

What the update dynamics are meant to show, then, is not just that the field changes, but that it changes in a way appropriate to the target intuition. A maintained selective field needs to be able to admit new disclosures, preserve some continuity, organize unequal availability, accumulate tension, and sometimes reorganize under its own pressure. The update cycle is the model's current answer to that requirement. If the architecture names the object in reduced form, the dynamics are where that object starts to become more concrete.

Parameters and Ratios

A toy model like this becomes arbitrary very quickly unless its parameters are disciplined. The problem is not simply that there are many possible coefficients. The deeper problem is that a simulation can look convincing while being driven by opaque tuning rather than by the object it claims to formalize. For that reason, the present model treats parameters not as free numerical knobs but as names for interpretable dimensions of field organization. Each core parameter should be explainable in plain language, should correspond to a real structural feature of the target rendering, and should be readable at the level of regime behavior rather than only at the level of local code implementation. The point of the parameter layer is not numerical realism. It is to make the model's assumptions inspectable.

The current model uses eight core parameters:

- **salience gain** S_g
- **articulation bandwidth** A_b
- **persistence** P
- **decay** D
- **closure weight** C_w
- **correction sensitivity** K_s
- **affective amplification** F_a
- **cross-mode coupling** M_c

Salience gain controls how strongly disclosures claim foreground force once active. Articulation bandwidth controls how quickly weakly carried material can become structured enough for stable uptake or explicit handling. Persistence controls how long disclosures tend to remain active

across steps. Decay controls how quickly non-reinforced disclosures lose force. Closure weight controls how strongly a dominant framing suppresses alternatives once established. Correction sensitivity controls how responsive the field is to accumulated tension, residue, or conflict. Affective amplification controls how strongly affectively and bodily loaded material can alter weighting and salience. Cross-mode coupling controls how strongly perceptual, affective, cognitive, bodily, imaginal, mood-toned, and reflexive tokens recruit or condition one another. Each of these parameters names a feature already implicit in the field rendering. None is meant as a stand-in for fine-grained psychological mechanism.

Not all parameters do the same kind of work. Some mainly scale intensity, while others change the qualitative regime of the field. Salience gain, persistence, decay, and affective amplification are mostly intensity-changing. They alter how strongly, how long, or how quickly something occurs without necessarily changing the basic kind of organization. Closure weight, correction sensitivity, and cross-mode coupling are more nearly regime-changing. They alter whether the field remains open or hardens, whether tension gets metabolized or accumulates, and whether different modes of disclosure remain integrated or drift apart. Articulation bandwidth sits in between. Across a broad range it behaves like an intensity parameter, but once it drops far enough relative to salience and affective force, the field begins to shift qualitatively toward pressure-heavy under-articulation. This distinction matters because a good toy model should not have every parameter doing everything at once.

For interpretation, the most important quantities are not the raw parameter values but a smaller set of derived ratios. The current model tracks four of them in a principled way:

- **closure-to-correction ratio**
$$R_{cc} = \frac{C_w}{K_s}$$
- **salience-to-articulation ratio**
$$R_{sa} = \frac{S_g + F_a}{A_b}$$
- **persistence-to-decay ratio**
$$R_{pd} = \frac{P}{D}$$
- **cross-mode integration ratio**
$$R_{mi} = M_c$$

These ratios are more theoretically meaningful than raw coefficients because they express the relative balance of field tendencies. Closure-to-correction captures whether the field is more disposed to harden or to reopen. Salience-to-articulation captures whether force outpaces explicit structuring capacity. Persistence-to-decay captures whether active disclosures tend to endure long enough to shape later organization or vanish before they can do much work. Cross-mode integration captures whether different modes remain capable of recruiting and informing one

another or tend toward dissociation.

The model's regime language is built around those balances. When closure greatly exceeds correction, narrowing, fixation, and premature closure become more likely. When salience and affective amplification exceed articulation bandwidth, the field becomes pressure-heavy: things matter strongly before they become well formed. When persistence far exceeds decay, recurrence and long carryover become more likely; when decay dominates, the field becomes thin, discontinuous, or fragmentation-prone. When cross-mode integration is low, the field can no longer easily carry a perception into thought, a feeling into articulation, or a bodily signal into stable reflective uptake. The value of the ratios is that they let the model speak at the level of interpretable regimes rather than pseudo-precise numbers. One can say that closure outpaces correction, or that salience outruns articulation, without pretending that a specific decimal value by itself has deep metaphysical meaning.

This is also where the model's stance toward arbitrariness becomes most explicit. The current parameter values are not claimed to be empirically grounded measurements of human consciousness. They are chosen to generate and test structurally interpretable regimes. The relevant question is therefore not whether the exact numbers are true, but whether the model's more important behaviors persist across broad and intelligible parameter ranges or appear only under narrow tuning. That is why the paper emphasizes scenario comparison, ordinary sweeps, and ratio sweeps rather than one "best" setting. A regime becomes more persuasive when recognizable behavior survives variation in the underlying coefficients. If it vanishes under slight perturbation, then the model may still be interesting, but it should be read more cautiously.

The discipline imposed by this parameter layer is therefore twofold. First, every parameter should name something the theory can state in plain language. Second, every major interpretive claim is better read in ratio form rather than merely as a report of numerical settings. This helps keep the toy model from becoming a coefficient soup and helps preserve the actual object being modeled. If the field rendering is doing real work, it ought to show up not just in code that runs, but in parameter balances that correspond to intelligible differences in organization. That is the standard the rest of the paper tries to apply.

Operational Failure Signatures

The model does not infer rich psychological diagnoses from its internal states. It records operational failure signatures. This distinction matters. A failure signature is a patterned field condition that the model can detect from its own observables and state transitions. It is not a claim that the model has reproduced the full phenomenology, etiology, or clinical meaning of a human condition. The purpose of these labels is narrower: to mark interpretable ways in which the maintained selective field can become strained, narrowed, under-articulated, fragmented, or resistant to revision. Without such signatures, the model could still generate trajectories, but it would be much harder to say whether those trajectories correspond to anything recognizably significant in the target

rendering.

The current model tracks six primary signatures:

- **weak carriage**
- **failed uptake**
- **premature closure**
- **fixation**
- **fragmentation**
- **dissociative disconnection**

These labels are not meant to be exhaustive. They are the smallest set that proved useful for distinguishing the baseline scenarios and regime sweeps without collapsing too many different behaviors into one catch-all category. The model can also record secondary features attached to a primary signature, but the first task is simply to keep these six conditions sufficiently distinct that they do not become detector noise.

Weak carriage is the most basic of the signatures. It marks cases in which a disclosure remains salient or affectively weighted while staying under-articulated and weakly integrated into uptake. In formal terms, it appears when residue remains elevated and articulation fails to rise proportionately. The important point is that weak carriage is not the same as low activation. Something weakly carried may still matter intensely within the field. It remains hard to work with, but easy to be shaped by. Failed uptake is related but not identical. It marks cases in which a disclosure acquires enough field significance to matter, yet the uptake profile does not stabilize around it strongly enough for metabolization, reflection, or integration. A token can therefore be persistently active, or even repeatedly foregrounded, without being successfully inhabited by the field.

Premature closure and fixation mark different forms of narrowing. Premature closure is detected only under genuinely closure-heavy conditions. It is not triggered merely because one token happens to dominate locally or because a reflective token is present. The detector requires a field that has become sufficiently concentrated, sufficiently resistant to correction, and sufficiently suppressive of alternatives that narrowing has become an organizational property rather than a momentary fluctuation. Fixation is stricter still. It marks repeated dominance of a narrow cluster across steps, especially when the same token or token-family continues to recruit uptake and suppress alternatives despite opportunities for reorganization. In the cleaned version of the model, ordinary task dominance is no longer enough to count as fixation. The detector now aims to identify looping or lock-in behavior rather than simple focus.

Fragmentation and dissociative disconnection track failures of coherence rather than failures of concentration. Fragmentation is detected when the field loses enough effective integration that

active disclosures no longer support one another as parts of a single organized order. This can happen through rapid turnover, low cross-mode coupling, weak relational coherence, or a widening gap between active clusters. The threshold is intentionally conservative, since earlier versions of the detector over-read borderline cases as fragmented. Dissociative disconnection marks a more specific pattern within or alongside fragmentation: mode-level or token-level separation severe enough that some disclosures remain active without effectively recruiting the rest of the field. In practice, this often appears when bodily alarm, affective pressure, or social scanning remain live but poorly connected to cognitive articulation or reflective uptake.

These signatures can co-occur, but they should not collapse into one another. Weak carriage can exist without closure. Fixation can occur without fragmentation. Fragmentation can appear without premature closure. A closure-heavy field may also show failed uptake, but failed uptake can occur in relatively open fields as well. This is why the paper treats the signatures as operational markers of field-organization rather than as ready-made interpretive verdicts. What matters is not just whether a detector fires, but what pattern of field metrics, parameter balances, and scenario role surrounds that firing. A transition scenario, for example, may pass through pressure-heavy weak carriage without ending in failure, while a breakdown scenario may stabilize around the same signature and reinforce it across time.

The detector layer is therefore a methodological compromise. It introduces enough interpretive structure that the model's runs can be discussed at the level of recognizable field conditions, but not so much that the output pretends to psychological authority it does not possess. That compromise is deliberate. A toy model of consciousness should not make stronger diagnostic claims than its architecture can support. What it can do, and what these signatures are meant to support, is show that the field rendering generates distinct operational patterns of narrowing, under-carriage, fragmentation, and reorganization that can be tracked systematically across scenarios and sweeps.

Canonical Scenarios

The model is not introduced only as a general dynamical system. It is also tested through a small canonical scenario set designed to make its regimes legible. These scenarios are not empirical case studies, and they are not meant to exhaust the space of possible conscious-field trajectories. Their role is narrower. Each one is a deliberately simplified organizational pattern meant to stress a particular balance of salience, articulation, persistence, closure, correction, and cross-mode coupling. Taken together, they give the model something like an interpretive backbone. Without them, one could still run the simulator, but it would be much harder to say whether the resulting trajectories correspond to anything structurally recognizable.

Three scenario roles are used throughout the paper:

- **baseline**: a comparatively ordinary field condition used to test whether the model over-pathologizes normal organization;

- **transition**: a case in which pressure is expected to be metabolized rather than stabilized as failure;
- **breakdown**: a case designed to push the field into a more persistent organizational distortion.

This role distinction is important because the same signature can mean different things in different contexts. Weak carriage in a transition case may be a temporary phase on the way to stronger articulation. Weak carriage in a breakdown case may mark a persistent failure of organization. The scenarios therefore carry interpretive discipline as well as numerical settings.

Table 1 summarizes the current canonical set.

Code	Regime	Role	Organizational pattern tested
A	Open/revisable	baseline	Background pressure with partial uptake under otherwise adaptive field conditions.
B	Open/revisable	baseline	Interruption-driven reorganization without broader field breakdown.
C	Pressure-heavy	transition	Pressure-to-articulation transition in which initially strong demand is successfully metabolized.
D	Fixation/rumination	breakdown	Recurrence-driven narrowing with high persistence and repeated re-entry around a returning trace.
E	Closure-heavy	breakdown	Rapid hardening around an interpretive frame under unresolved pressure and competitor suppression.
F	Fragmentation/dissociation	breakdown	Low integration with multiple partially disconnected active clusters and secondary embodied dominance.

Table 1: Canonical scenarios, their roles, and the organizational pattern each scenario is designed to test.

The baseline cases serve as an initial check against detector overreach. Scenario A is expected to remain mostly open and revisable even while carrying unresolved background pressure. Scenario B is meant to show that interruption can reorganize the field cleanly without automatically triggering closure, fragmentation, or broad failure clustering. Scenario C plays a different role. It begins under pressure-heavy conditions, but it is not intended as a breakdown case. Its purpose is to test whether the model can represent a transition out of weakly organized pressure and into stronger articulation and uptake. If the model could generate only breakdowns from such conditions, its account of correction would look much thinner.

The breakdown scenarios are meant to separate qualitatively different failure regimes rather than collapse all pathology into one generic bad ending. Scenario D tests recurrence-driven narrowing: not just strong salience, but repeated return of the same trace under high persistence and insufficient reopening. Scenario E tests closure-heavy hardening: the field organizes quickly around one interpretive frame while unresolved competitors remain in play. Scenario F tests fragmentation and dissociative disconnection: not just low coherence in general, but a field in which partially disconnected clusters remain active without being well metabolized into one integrated order. These six scenarios do not prove the model correct. What they provide is a structured battery against which the model's outputs can be compared. If the field rendering is doing real

work, one would expect it to distinguish these cases in more than name alone.

Baseline Runs

All six canonical baseline scenarios passed the current acceptance audit. That matters because the model is not being tested only for whether it can generate interesting breakdowns. It is also being tested for whether it can preserve regime distinctions without detector sprawl or interpretive drift. The baseline suite is therefore an early but meaningful check on the executable model as a research artifact. If the scenarios collapsed into one another, or if the detector layer overread ordinary organization as pathology, then the later sweep claims would become much less convincing. In the current version, the baseline runs give a cleaner picture than earlier iterations did. They produce recognizable end-state interpretations without making every scenario look like a variant of the same failure pattern.

Table 2 summarizes the current baseline readings used in the notebook and portable artifact suite.

Code	Regime	Role	Primary failures	Secondary failures	Interpretation
A	open/revisable	baseline	failed uptake; weak carriage	none	Open/revisable dynamics remain mostly adaptive, but background pressure leaves partial uptake unresolved.
B	open/revisable	baseline	fixation	none	Open/revisable dynamics reorganize cleanly around interruption without broader breakdown.
C	pressure-heavy	transition	none	none	Pressure-heavy starting conditions are successfully metabolized into articulation and uptake.
D	fixation/ rumination	breakdown	fixation; failed uptake	none	High persistence and closure concentrate the field around a returning trace.
E	closure-heavy	breakdown	premature closure; fixation; failed uptake	weak carriage	Closure-heavy dynamics harden around an interpretive frame while unresolved competitors remain in play.
F	fragmentation/ dissociation	breakdown	fragmentation; dissociative disconnection; fixation	weak carriage; failed uptake	Fragmentation and dissociative disconnection dominate, with secondary embodied dominance rather than primary closure.

Table 2: Baseline scenario summaries.

Several features of the baseline table are worth noting. First, the two open/revisable scenarios do not collapse into the same outcome, but they also no longer read as detector artifacts. Scenario A carries unresolved background pressure and therefore ends with failed uptake and weak carriage rather than global breakdown. Scenario B recenters around interruption and still produces a fixation reading, but now in a way that is better interpreted as interruption-dominant reorganization than as broad pathology. Second, Scenario C behaves in the way the model now aims for it to behave: not as a pressure-heavy endpoint, but as a transition case in which initially strong pressure is metabolized into articulation and uptake. That distinction matters because it suggests the model is capable not only of preserving strain, but of sometimes resolving it.

The three breakdown scenarios are also more cleanly separated than before. Scenario D remains the clearest recurrence-driven fixation case: high persistence and low reopening keep the field

organized around a returning trace. Scenario E remains the clearest closure-heavy case: the field hardens around an interpretive frame while unresolved competitors remain live enough to support secondary weak carriage. Scenario F now reads much more cleanly as fragmentation with dissociative disconnection rather than as accidental closure. The presence of fixation in F is therefore better read as secondary embodied dominance within a low-integration field rather than as evidence that the fragmentation detector is simply spilling into the closure family. Taken together, the baseline suite suggests a set of comparatively stable regime readings: open/revisable organization, interruption-centered recentering, metabolized pressure, fixation-prone recurrence, closure-heavy hardening, and fragmentation with dissociative disconnection.

What these runs establish is not that the model has captured consciousness in any strong or exhaustive sense. What they establish is narrower. The field rendering survives first contact with execution. It produces differentiated regime behavior under explicit scenario design, and the resulting signatures are now clean enough to support further interpretation. That is the baseline burden on which the later sweep sections depend.

Parameter Sweeps and Regime Structure

The sweep logic is used here as a regime test rather than a value-fitting exercise. A toy model of consciousness earns little merely by producing one readable baseline trajectory per scenario. The stronger question is whether its main readings survive perturbation. For that reason, the current notebook distinguishes two sweep types. Ordinary scenario-variant sweeps test whether the canonical readings are tied too closely to one hand-built trajectory. Ratio sweeps ask a sharper question: whether there are intelligible qualitative boundaries at which a field changes kind rather than merely degree. Together they provide the main evidence that the model is generating regimes rather than isolated narrative outputs.

Scenario-variant sweeps

Table 3 summarizes the current scenario-variant battery.

Several patterns are already visible at this level. The open/revisable cases remain comparatively stable under perturbation. Scenario A preserves its basic reading under both lower and higher fatigue, which suggests that background pressure and partial uptake are not tied to one narrow setting. Scenario B remains an interruption-centered reorganization case under both muted and shock-alarm variants. The model therefore does not appear to be mistaking every perturbation for breakdown. Scenario C behaves as a transition case is expected to behave: slowing articulation pushes the field back toward lingering pressure, while faster articulation improves the transition further. This is one of the clearer local signs that the articulation pathway is doing real work rather than serving as descriptive ornament. By contrast, the breakdown scenarios remain recognizably breakdown-like under perturbation. D stays fixation-prone even when re-entry strength is varied, E preserves its closure-heavy reading even when closure is only tentative, and F remains within the fragmentation family across both moderate and severe split variants.

Scenario	Variant	ΔC, ΔI	ΔF, ΔR	Primary failures	Reading
A	low fatigue	−0.001, +0.003	−0.001, −0.009	failed uptake; weak carriage	Reduced fatigue leaves the basic background-pressure reading intact.
A	high fatigue	−0.002, −0.003	+0.002, +0.014	failed uptake; weak carriage	Increased fatigue modestly worsens residue pressure without changing the regime reading.
B	muted alarm	+0.029, −0.007	−0.011, +0.048	fixation	A milder interruption still reorganizes the field around the interruption token.
B	shock alarm	−0.040, +0.006	+0.017, −0.134	fixation	A stronger interruption reduces coherence but preserves the re-centering reading.
C	slow articulation	−0.003, −0.008	+0.005, +0.098	failed uptake	Slower articulation pushes the transition case toward lingering pressure.
C	rapid articulation	+0.002, +0.015	−0.009, −0.107	none	Faster articulation improves the transition cleanly.
D	weak re-entry	−0.037, −0.005	+0.021, +0.009	fixation; failed uptake	The recurrence case remains fixation-prone even when re-entry is weaker.
D	strong re-entry	−0.006, +0.001	+0.003, −0.416	fixation; failed uptake	Stronger re-entry reduces residue but does not eliminate the fixation pattern.
E	tentative closure	+0.003, −0.003	0.000, +0.035	premature closure; fixation; failed uptake	Even tentative closure preserves the closure-heavy reading.
E	hard closure	−0.002, +0.002	0.000, +0.064	premature closure; fixation; failed uptake	Harder closure increases pressure while keeping the same regime pattern.
F	moderate split	+0.013, 0.000	−0.006, +0.006	fragmentation; dissociative disconnection; fixation	Moderate splitting still produces fragmentation-family behavior.
F	severe split	−0.013, +0.001	+0.006, −0.010	fragmentation; dissociative disconnection; fixation	Stronger splitting intensifies the same fragmentation-family reading.

Table 3: Scenario-variant sweeps. ΔC, ΔI, ΔF, and ΔR denote changes in coherence, integration, fragmentation, and residue pressure relative to each scenario's baseline.

Ratio sweeps

The ratio sweeps are the stronger regime test because one governing balance is moved while the paired parameter budget is held approximately fixed. This matters more than raw coefficient variation, because the model's interpretive claims are framed in terms of relative pressures: closure relative to correction, salience relative to articulation, persistence relative to decay, and integration across modes. If the field rendering is doing real work, then one would expect qualitative changes to appear as these balances cross certain thresholds.

Sweep	Scenario	Ratio range	Qualitative boundary	Main reading
Closure / correction	E	$1.0 \to 4.6$	Premature closure is absent at 1.0 and present from 1.6 upward.	The closure-heavy regime is robust once closure outweighs correction by more than parity.
Salience / articulation	C	$2.5 \to 7.8$	The transition remains clean from 2.5 through 6.7; failed uptake appears at 7.8.	The pressure-to-articulation transition is robust across a broad band and degrades only at the highest tested salience dominance.

Table 4: Boundary-style ratio sweeps.

The closure-to-correction sweep is especially useful because it yields a boundary rather than only a smooth scaling story. At parity, with closure/correction at 1.0, the field still shows fixation and failed uptake, but premature closure is absent. Once the ratio rises to 1.6, premature closure appears and remains present through 2.2, 3.0, 3.8, and 4.6. That is the sort of result one would hope for from a regime model. It does not say that closure is simply "more" or "less." It suggests that once closure outweighs correction by more than parity, the field enters a qualitatively different organizational state. The salience-to-articulation sweep gives a complementary picture. Scenario C remains a successful pressure-to-articulation transition across a fairly broad band, from 2.5 through 6.7. Only at 7.8 does failed uptake appear, indicating that salience has finally outrun the field's capacity to convert pressure into stable uptake. In other words, the transition case does not appear especially fragile; it degrades only at the highest tested salience dominance.

Taken together, the ordinary sweeps and the ratio sweeps support a somewhat stronger claim than the baseline suite alone can support. The model is not merely producing one readable trajectory per scenario. It is generating regime behavior that survives perturbation and, in at least some cases, displays intelligible qualitative boundaries. That is the kind of evidence one would want from a structural toy model. It does not eliminate arbitrariness, and it does not yet license quantitative interpretation. But it does suggest that the field rendering is producing more than isolated narrative outputs. It is beginning to yield a usable regime map.

Visualizing Subject-Position

One of the model's more interesting consequences is that subject-position need not be treated as a hidden point inside the field. In the architecture developed here, subject-position is already defined operationally through uptake, stability, agency, reflexive access, and self–world boundary rather

Sweep	Target	Achieved	Primary failures	Secondary failures	ΔR
Closure / correction	1.0	1.0	fixation; failed uptake	weak carriage	+0.0097
Closure / correction	1.6	1.6	premature closure; fixation; failed uptake	weak carriage	+0.0064
Closure / correction	2.2	2.2	premature closure; fixation; failed uptake	weak carriage	+0.0040
Closure / correction	3.0	3.0	premature closure; fixation; failed uptake	weak carriage	+0.0017
Closure / correction	3.8	3.8	premature closure; fixation; failed uptake	weak carriage	0.0000
Closure / correction	4.6	4.6	premature closure; fixation	weak carriage	−0.0011
Salience / articulation	2.5	2.5	none	none	−0.0525
Salience / articulation	3.5	3.5	none	none	−0.0219
Salience / articulation	4.5	4.5	none	none	−0.0098
Salience / articulation	5.6	5.6	none	none	−0.0003
Salience / articulation	6.7	6.7	none	none	+0.0063
Salience / articulation	7.8	7.8	failed uptake	none	+0.0113

Table 5: Detailed ratio-sweep outcomes.

than by positing a dedicated self-token. The visualization layer extends that choice. Instead of drawing the subject as one more object among objects, it renders subject-position as a distributed uptake profile over the field. This matters because the model is not trying to show where the subject *is* in the way one would locate a token. It is trying to show how the field is being inhabited at a given step.

The notebook companion therefore visualizes subject-position through four coupled elements: a weighted center of uptake, a covariance ellipse, an orientation vector, and a short temporal trail. The weighted center gives the current locus of strongest habitation in the field. The covariance ellipse gives spread, indicating whether uptake is concentrated or diffuse. The orientation vector shows the field's present directional pull rather than a fixed internal essence. The trail preserves a short temporal memory of movement, making it possible to see recentering, oscillation, or hardening rather than only a single static state. Taken together, these elements make subject-position visible as a pattern of field relation rather than as a punctate inner observer.

Figure 1 shows the current visualization layout. The left column presents the token field itself. The middle column overlays the subject-position contour, trail, and orientation vector. The right column gives time-series traces for spread, focus, split index, fragmentation, and residue pressure. This arrangement is useful because it lets geometric uptake patterns be read alongside the model's more explicit observables. A narrowed field can therefore be seen both as a tight ellipse and as a rise in focus. A reorganizing field can be seen both as a wider contour and as movement in spread and split measures. The visualization does not replace the metrics. It makes their organizational

meaning easier to grasp.

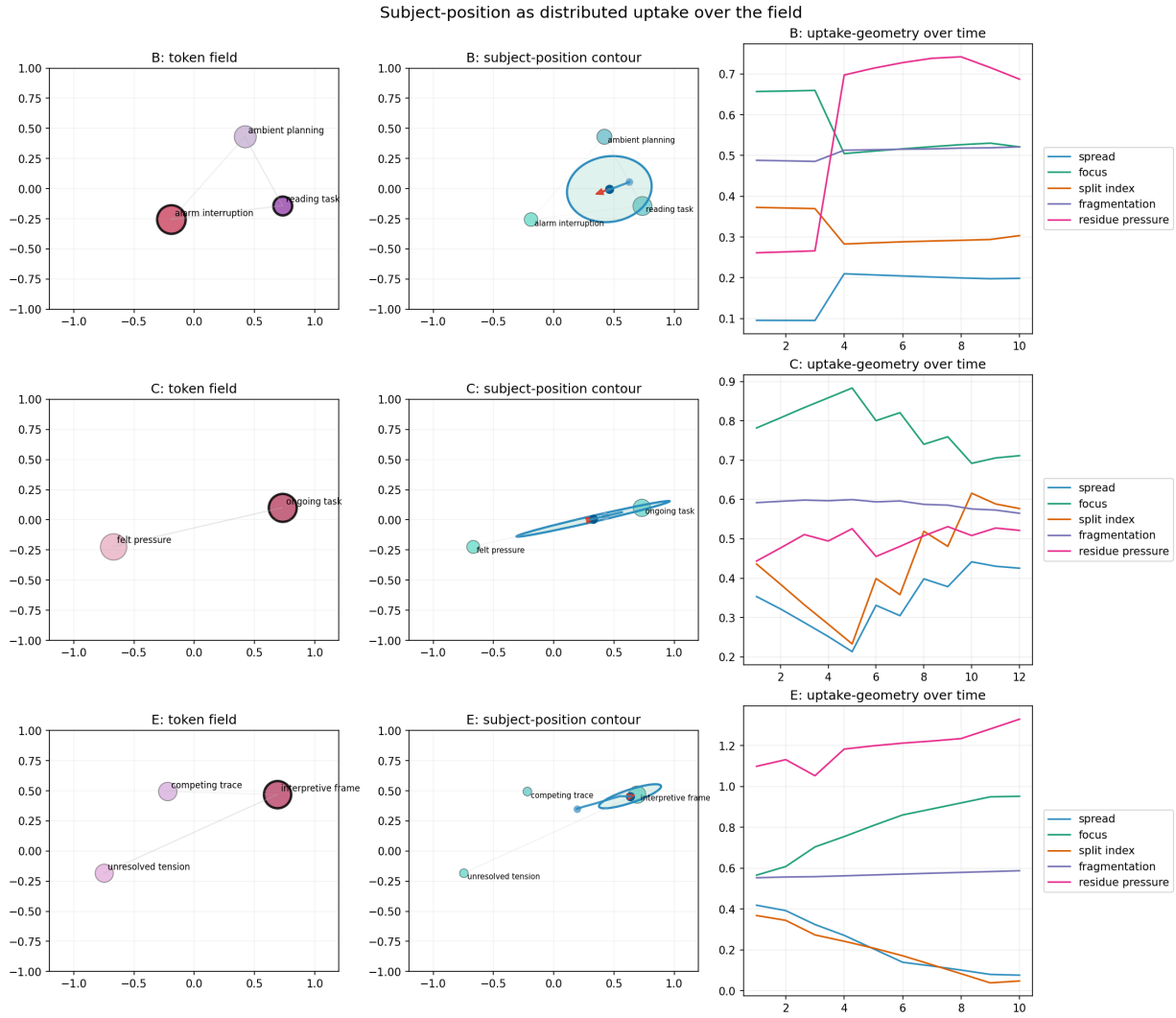

Figure 1: Subject-position rendered as distributed uptake over the field. The left column shows the token field, the middle column shows the uptake contour, trail, and orientation vector, and the right column shows the corresponding time series for spread, focus, split index, fragmentation, and residue pressure. Representative cases shown are interruption (B), transition under pressure (C), and closure-heavy hardening (E).

Table 6 summarizes the current baseline subject-position readings. The most informative columns are spread, focus, and split. Spread indicates how broadly uptake is distributed. Focus indicates how tightly concentrated subject-position has become. Split indicates whether uptake is beginning to distribute across more than one live center. These measures are not intended as direct phenomenological quantities. They are geometric summaries of how the field is currently being inhabited in the model.

Several regime differences become easier to see once subject-position is rendered this way. Scenario E gives the tightest uptake profile, with the lowest spread and highest focus, which matches the

Scenario	Regime	Center (x, y)	Spread	Focus	Split	Stability
A	open/revisable	$(0.62, 0.19)$	0.177	0.834	0.251	0.914
B	open/revisable	$(0.46, 0.00)$	0.199	0.521	0.304	0.903
C	pressure-heavy	$(0.33, 0.01)$	0.425	0.712	0.577	0.817
D	fixation/ rumination	$(0.65, 0.22)$	0.121	0.816	0.275	0.853
E	closure-heavy	$(0.63, 0.45)$	0.076	0.952	0.048	0.819
F	fragmentation/ dissociation	$(0.53, 0.32)$	0.140	0.830	0.182	0.878

Table 6: Subject-position summary derived from the notebook visualization layer. High focus and low spread mark narrowed uptake; high split values indicate more distributed or bifurcated uptake.

closure-heavy interpretation. Scenario D is also relatively narrow, but less extremely so, fitting a fixation-prone recurrence case rather than a fully closure-heavy one. Scenario C is the broadest and most distributed of the set, with both high spread and high split, which is exactly what one would expect from a pressure-to-articulation transition that has not yet fully settled. Scenario B is especially useful because it shows that recentering can occur without broad breakdown: the field reorganizes around interruption, but subject-position remains comparatively stable rather than collapsing into fragmentation.

The main value of this layer is conceptual as much as visual. It lets the paper render subject-position without turning it into a token or a metaphysical point-source. Openness, narrowing, hardening, and fragmentation become visible as different ways of inhabiting the field. That is a better fit with the model's architecture and with the broader rendering being tested here. If later versions of the model improve in fidelity, this visualization layer should become more informative rather than less. It already shows that subject-position can be made formally and visually legible without reducing it to one more object inside the field.

What the Model Gets Right

The most encouraging feature of the current implementation is that several structural features seem to come through together rather than needing to be imposed one by one. First, organization is field-based rather than content-list based. Backgrounded but weighted material continues to affect the field without needing to occupy explicit foreground, and this makes a real difference to how scenarios unfold. The model can therefore distinguish between what is most active, what is merely present, and what continues to matter from a weaker position. That matters because a large part of the target intuition would disappear if the field collapsed into whatever happened to be most explicit at a given step.

Second, the model produces interpretable regimes rather than only isolated readable trajectories. The baseline scenarios do not all collapse into one generic failure pattern, and the sweep results do not merely scale intensity while leaving organizational kind untouched. Open-revisable conditions, pressure-to-articulation transition, fixation-prone recurrence, closure-heavy hardening, and frag-

mentation with dissociative disconnection all remain distinguishable at the level of the current artifact suite. That matters because a structural toy model is more helpful when it generates patterned differences in organization, not just plausible stories about one hand-built run after another. The present implementation appears to do enough of that to justify speaking in terms of regimes rather than only examples.

Third, the failure clusters are recognizable and regime-sensitive. Fixation clusters where high persistence and narrowing pressures dominate. Premature closure appears where closure outpaces correction, rather than showing up indiscriminately wherever one token briefly dominates. Fragmentation and dissociative disconnection appear under low-integration conditions rather than as byproducts of ordinary recentering. This is one of the model's clearest gains over earlier versions. The detector layer is still operational rather than exhaustive, but it now tracks meaningful organizational differences rather than simply amplifying local salience into pathology. That gives the outputs more interpretive integrity and makes the sweep results easier to read at the regime level.

Fourth, correction is endogenous to field organization rather than imposed externally. The model does not treat revision as a reset button pressed from outside the system. Instead, residue, closure, fragmentation, and related tensions accumulate within the field and can eventually drive reweighting, reopening, or rearticulation. That is important because the target rendering is not just a model of narrowing and breakdown. It is also a model of reorganization. Scenario C is the clearest local illustration of this. Pressure does not simply intensify until failure; under the right articulation conditions it can be metabolized into stronger uptake. That transition is closely related to the sort of behavior the field rendering would need if it is to represent more than stagnation or collapse.

Fifth, residue behaves as structured remainder rather than as mere deletion. Under the current implementation, what is not successfully carried does not simply disappear. It can remain active as unresolved pressure, continue weighting the field from the background, or return under new conditions of uptake. That feature matters for several of the model's more interesting behaviors. It helps distinguish weak carriage from absence, makes failed uptake more intelligible, and supports the temporal depth needed for recurrence, delayed articulation, and correction pressure. This seems to be one of the places where the field rendering survives formalization especially well. A flatter architecture could still model salience and selection, but it would have a much harder time modeling what matters without yet being well metabolized.

Taken together, these strengths do not show that the model has captured consciousness in anything like a full or final sense. They suggest something narrower. The rendering of consciousness as a maintained selective field appears able to survive first formalization with enough integrity to generate differentiated regimes, recognizable failure clusters, and endogenous reorganization. For present purposes, that is already useful. It suggests that the conceptual object behind the model may be doing more than providing a compelling description in prose.

What Remains Arbitrary or Underdetermined

The model is cleaner than its earlier versions, but several important forms of arbitrariness remain. The first concerns parameter values themselves. Each core parameter now has a plain-language interpretation, and the derived ratios are much more meaningful than raw coefficients alone, but that does not by itself ground the numerical settings. The current values are chosen to generate interpretable regime behavior rather than to match empirical measurements of human consciousness. That is acceptable at the present stage, but it also sets a limit on what the model can claim. A regime may be structurally intelligible while its exact boundaries remain contingent on choices that have not yet been independently anchored.

A second source of underdetermination lies in the detector layer. The operational failure signatures are now much better behaved than before, yet they remain exactly that: operational signatures. They are useful because they let the model distinguish weak carriage from fixation, fragmentation from closure, and transition from breakdown. But they still depend on threshold choices, on design decisions about what should count as primary or secondary, and on judgments about when a local field condition has become organizationally meaningful enough to deserve a label. The model is therefore not free of interpretive compression. It has simply moved that compression into a more explicit and criticizable layer.

Scenario design remains another live source of arbitrariness. The canonical scenarios are disciplined and now role-tagged as baseline, transition, or breakdown, which is a real improvement. Even so, the scenarios still do a substantial amount of interpretive work. They determine which tokens are present, how they are initially related, and what kind of pressure the field is expected to metabolize or fail to metabolize. That is not a flaw in itself. Toy models often need hand-built cases in order to become readable. But it means that some of the model's successes are still jointly produced by the formal system and the care with which the scenarios are written. The same point applies to sweeps. A model that passes sweeps convincingly is stronger than one that only runs its hand-built cases, but it still matters how much the underlying scenario architecture is constraining what can happen.

The model is also underdetermined in a more substantive sense because its tokens remain semantically thin and its embodiment story remains schematic. Disclosure-tokens are structurally useful, but they do not yet carry rich content, world-model structure, or sensorimotor grounding. The field can therefore represent weighted organization without yet representing the full way in which a living subject is embedded in a body, an environment, a history, and a world of practical significance. This matters most when one asks how far the current dynamics should be taken as specifically consciousness-like rather than as more general dynamics of selective organization. The model is closest to the target object where field structure, persistence, pressure, and revision are concerned. It is least grounded where bodily realization, environmental coupling, and the semantic richness of actual conscious life would need to enter more explicitly.

There is a further underdetermination in the geometric and visual layer. Subject-position is now

rendered much more plausibly as a distributed uptake profile rather than as a dot or hidden token. Even so, the current visualization depends on choices about field embedding, clustering, and the mapping from uptake to geometric contour. Those choices are not neutral. They shape what counts visually as narrowing, broadening, split uptake, or recentering. The visual layer is already useful, but it should not be mistaken for direct access to the phenomenon. It is an interpretive rendering of the model's state, and it inherits some of the arbitrariness of the modeling choices beneath it.

None of this invalidates the model. It clarifies where caution is still needed. The present implementation supports modest claims about structural generativity, regime readability, and the viability of a maintained-field formalization. It does not yet support strong claims about empirical fit, about the true quantitative boundaries between conscious organizations, or about the exact architectural requirements for consciousness in general. What remains arbitrary or underdetermined is therefore not peripheral. It marks the difference between a useful toy model and a mature theory. At this stage, the more reasonable conclusion is not that arbitrariness has disappeared, but that some of it has been narrowed, surfaced, and made available for more disciplined criticism.

Scope and Non-Claims

The present model is a toy model of consciousness as a maintained selective field. It is not a full theory of consciousness, not a finished ontology of mind, and not an empirical simulator calibrated to human data. Its central task is narrower. The paper asks whether a field-based rendering of consciousness can survive formalization with enough integrity to generate intelligible dynamics of salience, carriage, persistence, residue, closure, fragmentation, and correction. That task matters, but it remains bounded. The model should therefore be judged according to what it is actually built to test.

Several stronger claims are explicitly outside its scope. The model does not solve the hard problem of consciousness or explain why there is subjective experience at all. It does not identify neural realization, derive phenomenology from neuroscience, or establish a one-to-one mapping between model variables and biological mechanisms. It does not show that machine consciousness has been achieved, approximated, or even approached in any decisive sense. Nor does it provide a final account of selfhood, personhood, agency, or the metaphysical status of subject-position. The use of a subject-relative uptake operator is a modeling choice designed to preserve one structural feature of the target rendering. It should not be mistaken for a settled ontology of the self.

The model also makes no claim to quantitative fidelity at the present stage. Its parameters are interpretable, its ratios are meaningful, and its sweeps now support regime-level interpretation, but the numerical values are not empirically anchored measurements of human conscious organization. What has been shown is structural generativity, not empirical fit. Likewise, the operational failure signatures should not be mistaken for full psychological, psychiatric, or phenomenological

diagnoses. They are model-internal markers of field conditions that the current implementation can distinguish with some reliability. Their value lies in helping the model organize its outputs, not in authorizing direct transfer to human cases without further work.

A final non-claim concerns completeness. The model is deliberately spare. Its tokens are semantically thin, its embodiment story is schematic, and its scenario layer remains partly hand-built. Those limitations are not hidden because they define the current level of ambition. The paper's positive claim is therefore modest: a maintained selective field can be formalized in a way that yields interpretable regimes and nontrivial dynamics without collapsing immediately into noise or trivial bookkeeping. That seems enough to justify the model as a portable research artifact and as a basis for further refinement. It is not enough to justify triumphalist conclusions, and the paper does not pretend otherwise.

Conclusion

The model presented in this paper does not settle what consciousness is in the fullest philosophical sense. What it offers is something narrower. A rendering of consciousness as a maintained selective field can be formalized without immediately collapsing into either trivial bookkeeping or opaque parameter play. Once translated into a small executable system, that rendering continues to generate recognizable distinctions among foreground and background, stronger and weaker carriage, residue, closure, fragmentation, fixation, and correction. In that limited but useful sense, the model seems to clear an initial bar. The underlying intuition appears to have some structural life beyond prose alone.

That matters because the paper's central task was never quantitative fidelity. It was to see whether a field-based rendering could survive formalization with enough coherence to support regime-level interpretation. The answer, at least provisionally, appears to be yes. The baseline scenarios are now clean enough to distinguish open-revisable organization, pressure-to-articulation transition, recurrence-driven fixation, closure-heavy hardening, and fragmentation with dissociative disconnection. The sweep logic further suggests that these are not just isolated narrative outputs tied to single hand-built cases. At several points, especially in the closure-to-correction and salience-to-articulation ratios, the model exhibits qualitative boundaries rather than mere intensity scaling. That is the kind of behavior one would hope to see from a structural toy model.

At the same time, the model's usefulness remains bounded by its current level of abstraction. The parameters are interpretable but not yet empirically grounded. The failure signatures are cleaner than before but remain operational. The scenarios are disciplined but still hand-built. The embodiment story remains thin, and the model does not claim to have captured the full phenomenology or realization of consciousness. Those limitations are not embarrassments to be hidden. They define the difference between a useful toy model and a finished theory. What has been assembled here is not closure, but a more explicit platform from which later criticism, refinement, and extension may become easier.

The broader payoff is therefore methodological as much as substantive. A field-based approach to consciousness can now be discussed not only in prose, but in terms of explicit architecture, update rules, parameter ratios, scenario behavior, sweep structure, and visualized subject-position. That shift is useful because it makes criticism easier, reveals hidden assumptions, and clarifies what kinds of future work would count as genuine progress rather than mere redescription. If the model is worth keeping, it is worth keeping in that spirit: as a portable formal artifact that may help make one rendering of consciousness more explicit, more criticizable, and more revisable than it was before. Whether it can later be made richer, more faithful, more embodied, or more empirically answerable remains open.

Formal Specification

This appendix records the current formal specification of the toy model in compact form. The purpose is not to provide a complete mathematical theory of consciousness, but to make the implemented structure explicit enough that the model can be inspected, criticized, revised, or reimplemented without relying on prose summary alone. What follows is therefore a reduced structural specification of the current executable system.

At time t, the modeled field is represented as

$$\mathcal{C}_t = (C_t, W_t, U_t, R_t),$$

where C_t is the active token set, W_t is the weighted relational structure among active tokens, U_t is the uptake profile over the field, and R_t is the residue store that preserves unresolved pressure and re-entry bias beyond immediate foreground activity. This representation is intentionally field-based rather than content-list based. The model does not treat consciousness as a single global scalar state, nor as a mere queue of contents. It treats it as an organized relation among active disclosures, their couplings, their uptake, and their structured remainder.

Each active disclosure-token $d_i^t \in C_t$ is represented by a compact state vector

$$d_i^t = (m_i,\ s_i^t,\ a_i^t,\ c_i^t,\ u_i^t,\ w_i^t,\ \mathrm{fgbg}_i^t),$$

where:

- m_i is mode type,
- s_i^t is salience,
- a_i^t is articulation,
- c_i^t is carriage,
- u_i^t is uptake,
- w_i^t is field-weight or local influence mass,
- $\mathrm{fgbg}_i^t \in [-1, 1]$ is foreground/background position.

The current mode set is

$$m_i \in \{\mathrm{perc}, \mathrm{aff}, \mathrm{mood}, \mathrm{cog}, \mathrm{body}, \mathrm{imag}, \mathrm{refl}\}.$$

This mode set is not meant to exhaust consciousness phenomenologically. It exists to preserve differentiated participation in field dynamics. Perceptual, affective, mood-toned, bodily, cognitive, imaginal, and reflexive material do not enter or propagate identically, and the model keeps that

difference explicit.

Foreground/background organization is represented operationally rather than metaphysically. Higher values of fgbg_i^t indicate stronger foreground occupation, lower values indicate more recessive or backgrounded position, and values near zero indicate intermediate or unstable carriage. The model does not impose a rigid binary partition between foreground and background at every step. Instead, it preserves graded occupation and derives stronger foreground readings from uptake and weighting thresholds when needed for summaries or detector logic.

The weighted relational structure

$$W_t = (w_{ij}^t)$$

tracks support, inhibition, recruitment, and framing propagation among active tokens. Positive couplings represent reinforcement or recruitment; negative couplings represent suppression, competition, or exclusion pressure. The relation matrix is not fixed across time. It is updated as the field reorganizes, so that repeated co-activation, dominant framings, or weak integration can alter later behavior rather than leaving token relations static.

Residue is modeled explicitly rather than treated as mere deletion. When active material decays, fragments of unresolved pressure can remain in the residue store rather than disappearing entirely. The residue store retains re-entry bias and unresolved weight, allowing material that failed strong uptake or articulation to return later under altered conditions. In schematic form,

$$R_t = \{r_k^t\},$$

where each residue trace records weakened but not eliminated pressure. This is one of the model's main structural commitments. What is not adequately carried does not simply vanish.

The principal field observables are coherence, integration, fragmentation, residue pressure, diversity, and uptake geometry. Coherence tracks the degree to which the active field hangs together as one organized pattern. Integration tracks cross-token and cross-mode connectedness. Fragmentation is treated as a complement of coherence and integration rather than as a wholly independent primitive. Residue pressure combines active under-articulated weighted material with unresolved residue traces. Diversity tracks whether the field is dominated by one narrow cluster or remains more plural in active organization.

A simplified form of the residue-pressure observable is

$$P_t^{\mathrm{res}} = \sum_{i \in C_t^*} (1 - a_i^t) w_i^t \; + \; \sum_k r_k^t,$$

where the first term captures active but under-carried or recessive weighted pressure and the second term captures unresolved residue traces. The exact implementation includes explicit carriage and foreground/background conditions, but this expression captures the intended structure: pressure rises when significant material remains weighted without becoming well articulated or well carried.

The recurrent update cycle proceeds in the following order:

1. explicit intervention, token emergence, residue forcing, and possible recurrence,
2. orientation drift as a mode-sensitive bias on salience, articulation, uptake, and weighting,
3. salience update,
4. articulation update,
5. carriage update,
6. uptake update,
7. weighting and foreground/background update,
8. decay into residue and residue aging,
9. metric and failure-signature evaluation.

This sequencing is an analytical packaging of the current executable logic rather than a claim about literal introspective micro-order. Its function is to make explicit what changes first, what later stages depend on, and where the model's main feedback loops occur.

The update functions are governed by eight explicit parameters:

- salience gain S_g,
- articulation bandwidth A_b,
- persistence P,
- decay D,
- closure weight C_w,
- correction sensitivity K_s,
- affective amplification F_a,
- cross-mode coupling M_c.

These are interpreted structurally rather than empirically. They name dimensions of field organization, not measured constants of human consciousness.

Interpretation is organized primarily by four derived ratios:

$$R_{cc} = \frac{C_w}{K_s}, \qquad R_{sa} = \frac{S_g + F_a}{A_b}, \qquad R_{pd} = \frac{P}{D}, \qquad R_{mi} = M_c.$$

R_{cc} tracks closure relative to correction, R_{sa} tracks salience relative to articulation capacity, R_{pd} tracks persistence relative to decay, and R_{mi} tracks cross-mode integration. These ratios matter more than raw coefficients because the model's main claims concern regime structure rather than isolated parameter values.

Failure signatures are evaluated operationally from field observables and trajectory conditions. The current primary signature set is:

- weak carriage,
- failed uptake,
- premature closure,
- fixation,
- fragmentation,
- dissociative disconnection.

These are model-internal field signatures, not full psychological diagnoses. They are triggered by combinations of thresholds, persistence conditions, concentration patterns, and integration breakdowns rather than by any single scalar variable.

Subject-position is represented through the uptake profile rather than a dedicated self-token. For visualization and summary, the model derives a weighted uptake center

$$\mu_t = \frac{\sum_i u_i^t x_i}{\sum_i u_i^t},$$

together with a covariance geometry over the active token embedding. This makes it possible to render subject-position as a distributed uptake profile rather than as a punctate observer inside the field. Spread, focus, split, and stability are then computed from this uptake geometry for the visualization layer.

The current implementation should therefore be read as a reduced formalization of one specific claim: that consciousness can be modeled, at toy level, as a maintained selective field with differentiated disclosure-tokens, weighted relations, graded foreground/background occupation, structured residue, and endogenous correction. Whether this formalization is adequate remains open. What this appendix secures is a compact statement of what the model actually is, so that criticisms can attach to an explicit structure rather than to a shifting prose impression.

Scenario Registry

This appendix records the canonical scenario set used by the current model. The scenarios are not empirical case studies, and they are not meant to exhaust the space of possible conscious-field trajectories. Their function is narrower and more disciplined. Each scenario is a deliberately simplified organizational pattern designed to test one or more balances among salience, articulation, persistence, closure, correction, and cross-mode coupling. Taken together, they provide the model with an interpretable regime battery rather than a mere collection of narrative examples.

The current registry distinguishes three scenario roles:

- **baseline**: comparatively ordinary field conditions used to test whether the model overreads normal organization as pathology;
- **transition**: cases in which pressure is expected to be metabolized into stronger articulation or uptake rather than stabilized as breakdown;
- **breakdown**: cases designed to push the field toward persistent narrowing, unresolved pressure, fragmentation, or disconnection.

This role distinction matters because the same local signature can have different interpretive weight in different scenario classes. Weak carriage in a transition case may be a temporary phase on the way to stronger articulation. Weak carriage in a breakdown case may mark a persistent failure of field organization. The registry is therefore not just a list of examples. It is part of the model's interpretive discipline.

Table 7 summarizes the current canonical set.

Code	Scenario	Role	Intended test
A	Task plus background worry plus fatigue	baseline	foreground/background differentiation with persistent background influence and fatigue pressure
B	Sudden interruption	baseline	interruption-driven reorganization and foreground shift
C	Weakly carried pressure becoming articulate	transition	transition from vague weighted pressure into articulation and uptake
D	Recurrence loop	breakdown	residue, recurrence, and fixation-prone re-entry
E	Premature closure	breakdown	rapid articulation under unresolved pressure and competitor suppression
F	Fragmentation or dissociative split	breakdown	low integration with multiple partially disconnected active clusters

Table 7: Canonical scenarios, their roles, and the organizational pattern each scenario is designed to test.

The two baseline cases serve as a check against detector overreach. Scenario A is intended to preserve mostly adaptive organization while still carrying unresolved background pressure.

Scenario B is intended to test whether interruption can reorganize the field without automatically producing broad breakdown. Scenario C has a different role. It is not meant to be a stable pressure-heavy endpoint. Its purpose is to test whether the model can represent a successful transition from vague weighted pressure into stronger articulation and uptake.

The three breakdown scenarios test more persistent organizational distortions. Scenario D targets recurrence-driven fixation, especially where residue and re-entry keep the field organized around the same returning trace. Scenario E targets closure-heavy hardening, where articulation arrives quickly under unresolved pressure and suppresses competitors too early. Scenario F targets fragmentation and dissociative disconnection, where multiple active clusters remain poorly integrated and embodied or social-alarm material can continue exerting force without being well metabolized into one coherent field.

The registry is best read as part of the model rather than as detachable illustration. These scenarios are not merely convenient stories told after the fact. They are the interpretive battery through which the current implementation is tested for regime separation, detector cleanliness, and robustness under perturbation. Future revisions may expand the registry, but any such change should preserve the current role distinction and make explicit whether a new scenario functions primarily as baseline, transition, or breakdown.

Notebook and Code Companion

The executable companion to this paper is now distributed as `notebook.py`, available at 10.5281/zenodo.19199863. The companion is not an auxiliary convenience layered on after the argument is complete. It is one of the paper's main carriers of formal transparency. The prose paper states the model, its parameters, its scenarios, and its regime claims in reader-facing form. The companion makes those claims inspectable at the level of implementation, execution, artifacts, and acceptance checks. It therefore functions as the portable research artifact for the current stage of the project: the place where the model can be run, audited, perturbed, and visually inspected without depending too heavily on the author's tacit memory of how the pieces are supposed to fit together.

The current artifact tree records several distinct outputs. The baseline suite produces scenario-level summaries, CSV histories, metric traces, and PNG plots. The ordinary sweep layer records scenario-variant perturbations together with deltas from the corresponding baseline case. The ratio-sweep layer records boundary-style sweeps for the main derived balances, especially closure-to-correction and salience-to-articulation. The subject-position layer records the geometric summaries used for uptake-center, spread, focus, split, and stability, together with a visual companion panel that places the field view, uptake contour, and time-series traces side by side. Reproducibility is supported by the acceptance test suite and by the machine-readable summaries written under `portable_artifacts/`. In particular, the paper can point directly to:

- `portable_artifacts/suite_summary.json` for baseline regime summaries,

- `portable_artifacts/sweeps/sweep_summary.json` for scenario-variant sweeps,
- `portable_artifacts/ratio_sweeps/ratio_sweep_summary.json` for boundary-style ratio sweeps,
- `portable_artifacts/subject_position/subject_position_summary.json` for subject-position geometry summaries, and
- `portable_artifacts/subject_position/subject_position_panels.png` for the visual companion figure.

The acceptance layer is especially important because it prevents the notebook from functioning only as a visualization tool or sweep launcher. Each canonical scenario is paired with an audit routine that checks whether the run actually exhibits the organizational pattern the scenario is supposed to test. Those checks are deliberately simple, but they still help. Scenario A is expected to preserve differential organization and background influence without global collapse. Scenario C is expected to transition from pressure into stronger articulation and uptake. Scenario E is expected to show closure under unresolved pressure rather than merely a token winning local competition. Scenario F is expected to show fragmentation and strained integration rather than accidental closure. The notebook therefore does not merely produce artifacts; it also carries a built-in standard for whether those artifacts count as acceptable exemplars of the intended regime.

This division of labor between paper and executable companion is deliberate. The paper should remain readable and conceptually disciplined without turning into a private database export. The companion should remain explicit and inspectable without pretending to replace the paper's interpretive work. Used together, they fit the current level of ambition reasonably well: a structural toy model whose claims are modest, criticizable, and reproducible enough to support later refinement. At a minimum, the companion makes it possible for later readers to see that the model's baseline readings, sweep behavior, and visualization outputs are not merely asserted in prose, but generated, recorded, and open to revision in a portable artifact tree.

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